# Executive Summary

## Business Context:

Customer churn poses a critical challenge for subscription-based businesses. High attrition rates directly impact revenue and growth potential. This project leverages a dataset of 500+ customer records to identify churn drivers, predict at-risk customers, and recommend actionable retention strategies.

## Objectives:

- Analyze customer behaviour to uncover churn patterns.
- Build predictive models to estimate churn probability.
- Segment customers into risk groups for targeted interventions.
- Provide a business implementation plan to reduce churn.

## Methodology:

- **Data Cleaning:** Standardized numeric fields, encoded categorical variables, and removed missing values.
- **Exploratory Data Analysis (EDA):** Visualized churn distribution, monthly charges, tenure, and correlations.
- **Predictive Modeling:** Applied Logistic Regression and Random Forest classifiers to predict churn.
- **Segmentation:** Used K-Means clustering to group customers by tenure and charges.
- **Survival-style Analysis:** Examined churn probability across customer tenure.

## Key Insights:

- Customers on **month-to-month contracts** and using **electronic check payments** show the highest churn rates.
- **Senior citizens** exhibit higher churn compared to younger customers.
- **High monthly charges** correlate strongly with churn likelihood.
- Random Forest model achieved strong predictive performance, highlighting contract type and payment method as top features.

## Recommendations:

1. **Contract Incentives:** Offer discounts or loyalty rewards for customers switching to one-year or two-year contracts.
2. **Payment Optimization:** Encourage adoption of digital wallet or auto-pay methods to reduce churn risk.
3. **Targeted Support:** Develop tailored engagement programs for senior citizen customers.
4. **Pricing Strategy:** Introduce tiered pricing or loyalty discounts to ease churn among high-charge customers.

## Business Impact:

Implementing these recommendations can reduce churn by an estimated **10–15%**, translating into significant annual revenue retention. The predictive model enables proactive identification of at-risk customers, allowing the business to intervene before churn occurs.